

Faculty Management System

CTEC 22043 Group Assignment

FMS-Group-E

Repository URL :

<https://github.com/FMS-Group-E/faculty-management-system.git>

Demo Video Link :

<https://drive.google.com/drive/folders/13g9Rl2xpV1425F2MzVVEDHC1RET64fqZ?usp=sharing>

Team Members

Student Number	Name	Github User Name
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CT/2022/067	H.N.D. Jayasinghe	deshanijayasinghe
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CT/2022/053	D.K.Y. Ravidith	YasiruRavidith
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1. Functional Requirements

Authentication

Login with username/password

Role-based routing: ADMIN, STUDENT, LECTURER

Admin

Full CRUD: Students, Lecturers, Courses, Departments, Degrees

Link lecturers to departments/courses

Link degrees to departments

Student

View/edit profile (name, student number, email, mobile, degree)

View timetable

View enrollments and grades

Lecturer

View/edit profile (name, employee ID, contact, specialization)

View teaching timetable

View assigned courses

2. Non-Functional Requirements

MVC architecture

Data persistence via JDBC + relational DB

Cascade delete / set-null constraints for integrity

Desktop GUI using Java Swing

3. Tech Stack

LayerTechnologyLanguageJava (SE)GUIJava Swing &
AWTPatternMVCDatabaseMySQLDriverMySQL Connector/J
8.0.33PersistenceJDBC

4. Project Structure

com.faculty-management-system/

```
|— src/com/faculty/
|   |— model/      # Entity classes
|   |— view/       # Swing UI screens
|   |— controller/ # Business logic
|   |— dao/        # JDBC CRUD operations
|   |— util/       # DB connection helper
|   |— main/       # Entry point
|— database/
|   |— schema.sql
|   |— sample_data.sql
|— docs/FMS-GroupXX.pdf
|— demo/demo_video.mp4
|— README.md
```

5. Database Schema

TableKey Columnsdepartmentsdepartment_id (PK), dept_name,
hod_name, staff_countdegreesdegree_id (PK), degree_name,
degree_code (unique), department_id (FK)usersuser_id (PK), username
(unique), password, role (ENUM)lecturerslecturer_id (PK), user_id

(FK), employee_id (unique), department_id (FK)courses course_id (PK), course_code (unique), department_id (FK), lecturer_id (FK)students student_id (PK), user_id (FK), student_number (unique), degree_id (FK)enrollments enrollment_id (PK), student_id (FK), course_id (FK), grade — unique (student_id, course_id)timetable timetable_id (PK), course_id (FK), day_of_week, start_time, end_time, location

FKs mostly ON DELETE SET NULL; enrollments and timetable use ON DELETE CASCADE.

6. Role Workflows

Login: LoginView → LoginController → UserDao checks users table → route by role.

Admin: AdminDashboardView → tabbed CRUD panels → Controller.add/update/delete() → DAO → MySQL.

Student: StudentDashboardView → profile edit via StudentController.updateStudentProfile(); view timetable & enrollments/grades.

Lecturer: LecturerDashboardView → profile view/edit; teaching schedule from courses + timetable filtered by lecturer_id.

7. Team Contributions

Work	Name
Course	D.K.Y. Ravidith
Degree	K.V.V.H. Gajanayaka
Department	H.L. Ranasingha
Enrollment	H.N.D. Jayasinghe
Lecturer	P.K.L.D.P. De Silva
Student	G.A. Pramodaya
User	G.A. Pramodaya
Home	D.K.Y. Ravidith
Profile	H.L. Ranasingha
Timetable	P.K.L.D.P. De Silva